

FarmSetu

Digital Agriculture and Farmer Support Ecosystem

Full-Stack Project Summary & Status Report

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System Architecture, Features, and Implementations

1. Executive Summary

FarmSetu (built on the Agro-Tech Lumina design system) is a next-generation digital agricultural platform that bridges the gap between traditional farming and high-tech data insights. By combining the organic vitality of farming with the precision of modern SaaS, FarmSetu empowers farmers with real-time tools to manage crops, machinery, water scheduling, and labor requirements efficiently.

This report provides a comprehensive summary of the project architecture, full-stack tech stack, and implemented modules - focusing on recent additions: the Water Scheduler, Labor Booking Board, and Machinery Rental (Uber for Tractors) modules.

2. System Architecture and Tech Stack

The system is built as a split full-stack application (Decoupled Backend API + SPA Frontend), ensuring high scalability, modularity, and responsiveness.

Backend Architecture (Spring Boot)

- **Core Engine:** Java 17+, Spring Boot, Spring Data JPA, Hibernate ORM.
- **Database:** PostgreSQL database containing schemas for crops, orders, marketplace, labor, machinery, and water scheduling.
- **Security:** Token-based JWT authentication, customizable User Roles (FARMER, EXPERT, ADMIN, SELLER), and transactional security guards.
- **Notifications:** Transactional Event Notification service broadcasting UI alerts for status updates, bookings, and alerts.
- **WebSockets:** Real-time communication broker for instant chat and status queues.

Frontend Architecture (Angular)

- **Core Engine:** Angular 17+ single-page application built with standalone components, dependency injection, and RxJS reactive streams.
- **State Management:** Angular Signals for lightweight, reactive frontend state and notification count synchronization.
- **Internationalization:** Custom JSON-based i18n translation system supporting English (en.json) and Hindi (hi.json).
- **Styling:** Vanilla CSS / TailwindCSS applying the Agro-Tech Lumina design system (vibrant glassmorphism, rounded corners, dark mode aesthetics).

3. Core Platform Modules (Part 1)

A. Dashboard and Profile Workspace

The central command workspace where farmers view weather forecasts, crop pins, current marketplace cart statuses, and incoming notifications. Provides localized profile setup for coordinate tracking and village boundaries.

B. Mandi Bhaav and Finder (Market Trends)

- **Mandi Finder:** Allows radius-based search of local crop markets (Mandis) using geographic coordinates.
- **Watchlists:** Farmers pin specific crops and mandis to track daily min/max/modal prices.
- **ROI Calculator:** Transport Cost ROI Calculator. Calculates transport costs against mandi prices to identify the most profitable market.
- **Price Forecasts:** 15-day AI forecasting displaying price trends based on arrival volume and historical data.

C. Water Queue (Baari) Scheduler

- **Water Scheduling:** Digitized water slot booking, helping community members share common water sources (wells, canals, pumps).
- **Live Queue Timeline:** Visualizes active water flow, next in line, and waiting approval slots.
- **AI Weather Advisor:** Rainfall Alert System. If high rainfall is forecast on the requested day, the AI intercepts, issues warnings, and prompts rescheduling.

D. Crop Calendar and Crop Recommendation

- **Recommendations:** Recommends crops based on soil, district, and weather profiles.
- **Calendar Tasks:** Generates daily task reminders for watering, fertilizing, and pesticide applications for pinned crops.

4. Core Platform Modules (Part 2)

E. Labor Booking Module (Farm Gig Economy)

Solves peak-season labor shortages by introducing a village-level agricultural gig economy board.

- **Job Posting:** Farmers post bulk job listings (e.g. "Onion harvesting, 5 workers, wage: Rs. 400/day").
- **Applications:** Local laborers apply, submitting phone numbers and profiles.
- **Auto-Close Constraint:** Once hired workers reach the required worker count, the job status flips to FILLED and all remaining pending applications are automatically rejected with push notification alerts.

F. Machinery Rental Module (Uber for Tractors)

Peer-to-peer equipment rental system allowing heavy machinery owners (Tractors, Drones, Harvesters) to monetize idle equipment by renting to neighboring farmers on an hourly/daily basis.

- **Discovery:** Nearby active equipment is discovered using high-precision Haversine distance calculations, filtered by radius (5km to 100km) and categories.
- **Collision Prevention:** Schedule validation query (JPQL countOverlapping) prevents double-bookings. Booking approvals check for overlapping time-slots to guarantee schedules.
- **Cost Calculation:** Dynamic hours-to-days cost previews. If duration ≥ 8 hours, it applies the cheaper rate between daily (rounded up) and hourly, providing renters with transparent estimates.

G. Interactive Chat and Expert Support

- **Farm Chat:** WebSocket-powered community groups for farmers to share local updates.
- **Expert Consulting:** One-on-one secure consultations between farmers and certified agronomists.

H. Disease Detection and Marketplace

- **Disease Classifier:** Uploads photos of crop leaves to analyze diseases and receive treatment remedies.
- **Marketplace:** Comprehensive agro-input e-commerce platform for purchasing seeds, fertilizers, and tools. Supports ordering, bidding, and cart logistics.

5. Admin and Management Tools

A full-featured Admin Control Center enables administrators to manage systemic reference data and run platform-wide communications:

- **User Management:** View user details, role assignments, and verify credentials.
- **Market Uploads:** Bulk upload mandi prices using file parsers.
- **Broadcaster:** Email Broadcaster tool to send newsletters or weather alerts to user lists.
- **Water & Order Controls:** Monitor orders, delivery logistics, and override water queue scheduling disputes.

6. Core Technical Highlights

- **Overlap Logic:** JPQL query comparing time boundaries: ``start1 < end2 AND end1 > start2`` ensures schedule safety.
- **Haversine Discover:** Calculated dynamically in memory ($R = 6371 \text{ km}$) to enable lightweight nearby searches without PostGIS database overhead.
- **Unified i18n:** Synchronized translation pipes dynamically render all views in Hindi and English.

7. Future Roadmap

- **IoT Integration:** Real-time soil moisture sensors linking to the Water Scheduler for automated irrigation scheduling.
- **Payment Gateway:** UPI integration for instant marketplace checkouts, water scheduler fees, and machinery rental security deposits.
- **Clouinary Uplink:** Full backend integration with image upload services for equipment listing photos and leaf analysis.